



Dart Programming Assignment: Functions

Question 1: Cube Calculator

Write a standard function named `calculateCube` that takes one whole number (`int`) as a parameter. The function should calculate the cube of that number ($\text{number} \times \text{number} \times \text{number}$) and return the final answer as an integer.

Question 2: Text Cleaner

Write a function called `cleanAndUpper` that takes a string of text as a parameter. The function should:

- Strip away any extra blank trailing spaces from both ends of the text using `.trim()`.
- Convert the remaining text completely to uppercase letters using `.toUpperCase()`.
- Return the newly cleaned text string back to the caller.

Question 3: Optional Profile Data

Create a function named `displayUserProfile` that prints out a user's location profile. It must accept the following parameters:

- `firstName` (Positional Required)
- `lastName` (Positional Required)
- `country` (Positional Optional, with a default fallback value of "Nepal")

Test the function in `main()` to verify it can accept either two parameters or three parameters correctly.

Question 4: Secure Login Simulation

Create a function called `secureLogin` that simulates an app authorization credential check. It must use Named Parameters:

- `username` (Named Required)
- `password` (Named Required)
- `deviceType` (Named Optional, with a default value of "Mobile Phone")

The function should simply print a message inside its body matching this format:
"Attempting login for user [username] on a [deviceType]..."